

Energy band gap of intrinsic conductor

OBJECT:- To determine the energy band gap of a semi-conductor by measuring the variation of reverse saturation current with temperature.

APPARATUS: Four probe arrangement, thermometer, constant power supply, voltmeter

FORMULA/THEORY:-

The variation of reverse saturation current (I_s) with temperature (T) in case of p-n junction diode is given by

$$2.303 \log_{10} I_s = \text{Const.} - \frac{\Delta E}{kT} \dots\dots\dots(1)$$

Where I_s is in microampere, ΔE is the energy band gap of semi-conductor in electron volt and k = Boltz man constant = 1.38×10^{-23} J/K

Eq. (1) can be rewritten as

$$\log_{10} I_s = \text{another.Const.} - 5.036 \times \Delta E \times \frac{10^3}{T}$$

Now plot a curve between $\log_{10} I_s$ and $\frac{10^3}{T}$. It will be a straight line whose slope will

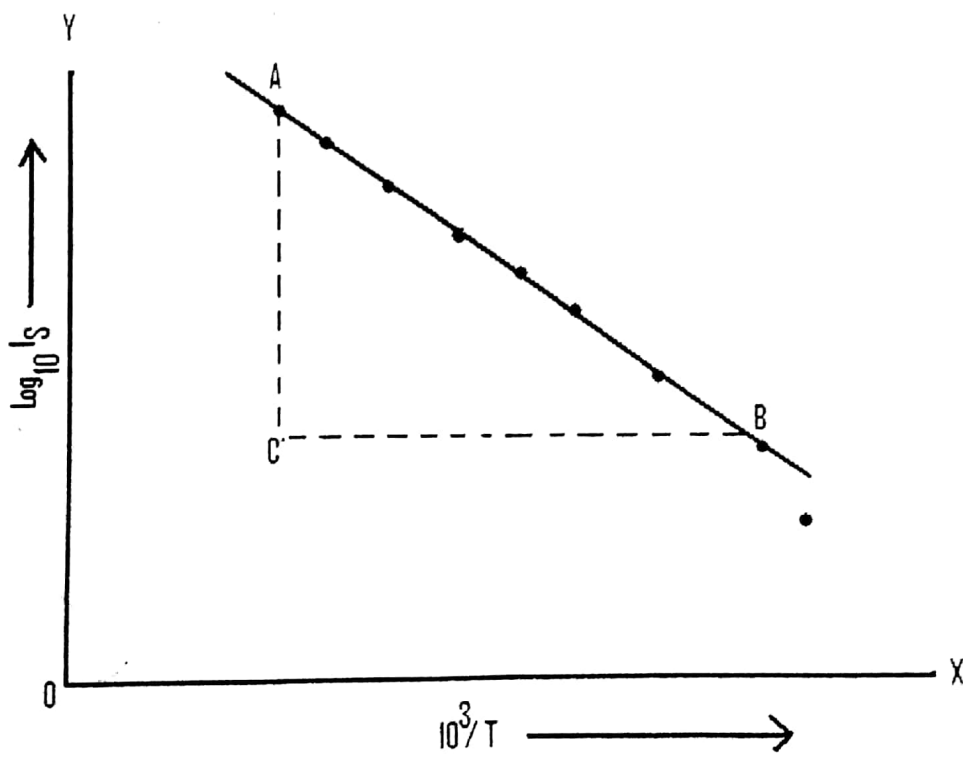
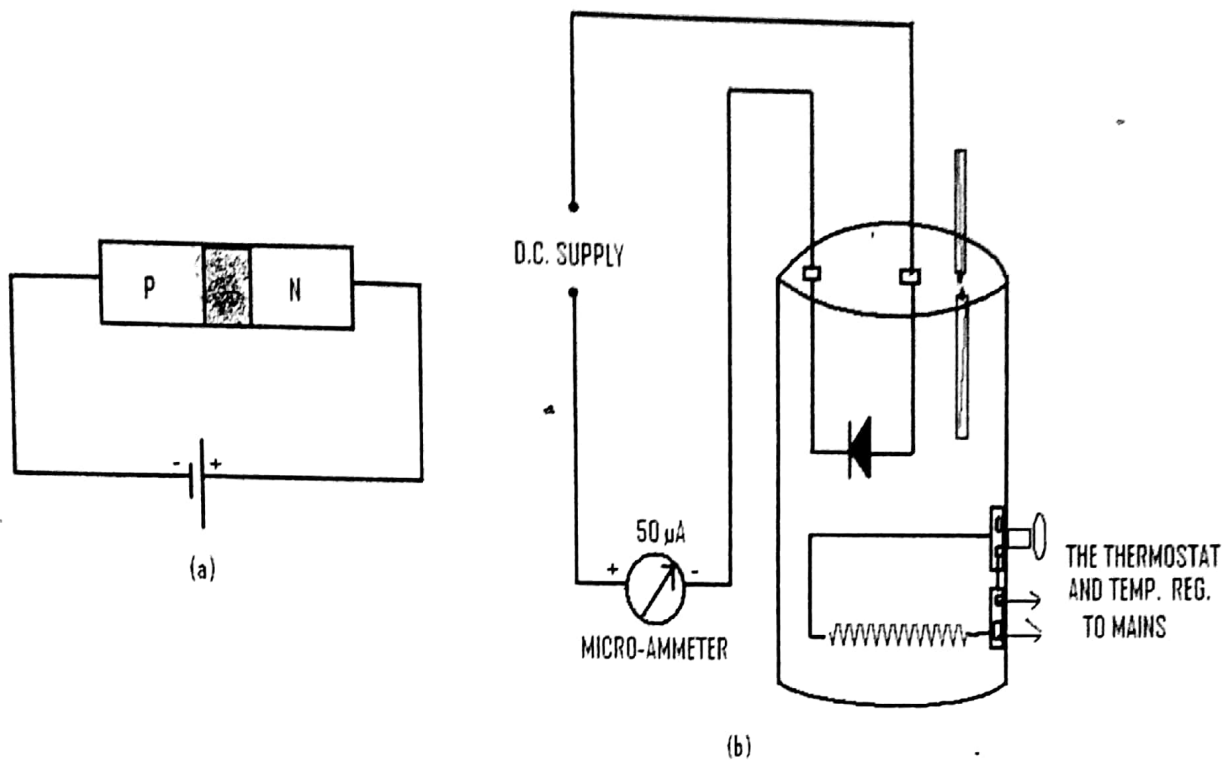
$$\text{be } \frac{\log_{10} I_s}{10^3/T}.$$

Thus slope of the line = $- 5.036 \Delta E$

or band gap of the semi-conductor $\Delta E = \frac{\text{Slope of line between } \log_{10} I_s \text{ and } 10^3/T}{5.036}$

5.036

DIAGRAM:-



PROCEDURE:-

1. A reverse biased p-n junction diode is shown in above figure (a). insert the p-n junction diode in the oven and connect its two leads to the terminals provided for the purpose. Connect the terminals through a $50\mu\text{A}$ D.C. micro-ammeter to a D.C. supply so that the diode is reverse biased. Insert a thermometer in its socket.
2. Connect the heating element of the oven to A.C. main supply. Switch on the mains and allow the temperature of the oven to rise to about 70°C - 75°C . Adjust the thermostat and temperature regulator so that the temperature remains steady at 70°C .
3. Note the temperature, adjust the reverse bias so that the micro-ammeter reads about $30\mu\text{A}$. Note the current reading.
4. Now switch off the mains supply, so that the oven cools slowly. As the temperature falls the micro-ammeter reading goes on decreasing. When the current decreases by $5\mu\text{A}$ note the current and the corresponding reading of temperature. Go on noting the current and corresponding temperature for every $5\mu\text{A}$ decrease in current in the diode circuit.

OBSERVATION TABLE:-

S. No.	Current I_s (μA)	Temperature ($^\circ\text{C}$)	Temperature T ($^\circ\text{K}$)	$10^3/T$	$\log_{10} I_s$
1.					
2.					
3.					
4.					
5.					

5. Taking $103/T$ along the X - axis and $\log_{10}I_s$ along the Y-axis, plot a graph between $\log_{10}I_s$ and $103/T$. The graph is a straight line as shown in above figure. Take two points A & B on the graph. From A draw a line parallel to Y-axis and from B a line parallel to X-axis meeting at C.

Then slope of the line AC/BC

$$\text{Band gap } \Delta E = \frac{\text{Slope}}{5.036} = eV$$

RESULT:-

PRECAUTIONS:-

1. Current through the sample should be kept constant.
2. Temperature should be taken carefully.
3. Semiconductor wafer being very brittle, so care must be taken in its mounting on the platform below the four probe arrangement.
4. Exert minimum pressure to get electrical contact of probe with the wafer.
5. A reverse biased p-n junction diode.
6. Reading of current must be taken when the temperature is decreasing.
7. Reading of current and temperature must be taken simultaneously.